

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460387.753 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.016
Transit depth (Rp/Rs)^2	3.72 +/- 0.6 [%]
Orbital Inclination (inc)	81.36 +/- 1.33
Airmass coefficient 1 (a1)	1.201 +/- 0.015
Airmass coefficient 2 (a2)	-0.132 +/- 0.07
Scatter in the residuals of the lightcurve fit is	1.22 %
Best Comparison Star	#1 - [370, 49]
Optimal Aperture	4.16
Optimal Annulus	12.45
Transit Duration (day)	0.0505 +/- 0.0051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.193 ± 0.016 vs 0.1594 (deviation 1.88σ)
Duration fit vs published	0.0505 d vs 0.0512 d (deviation 0.13σ)
Mid-time O–C	5.3 min
Depth SNR	10.8
Residual scatter	1.22 %

Light curve

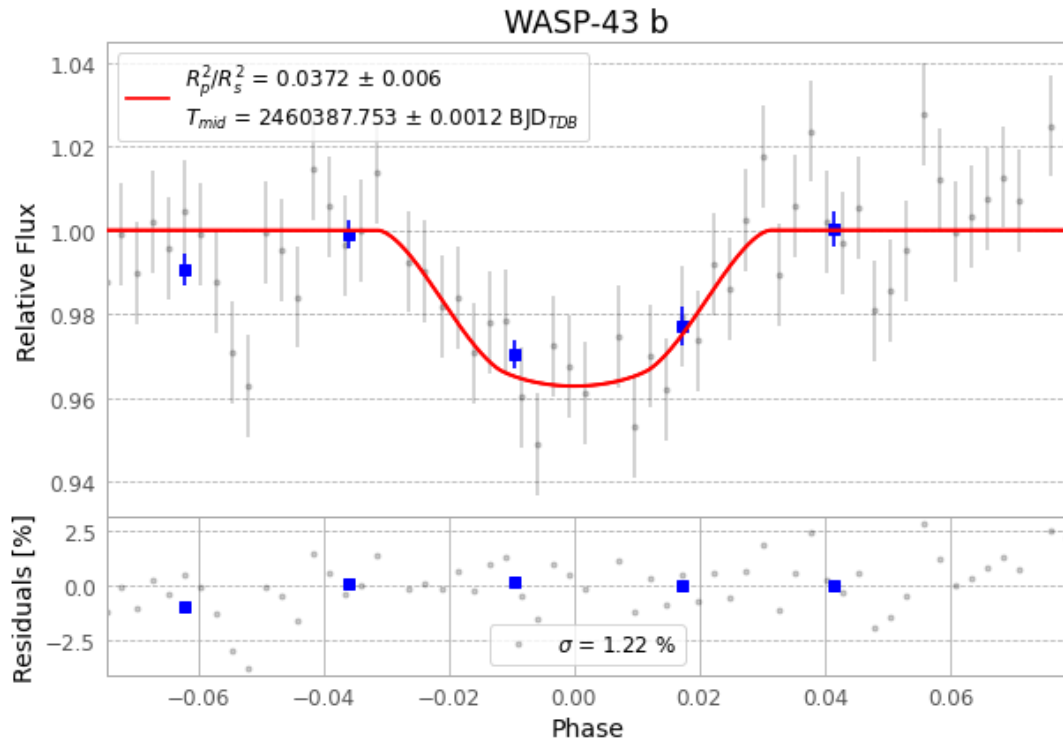


Figure 1 — FinalLightCurve_WASP-43 b_2024-03-17.png (SHA-256 0188072f85a74284...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [490, 231], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b5532181afe34c1e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

61 FITS frames (40 MB), WASP-43240318042746.FITS → WASP-43240318072715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-240318.log (1dda98c3474f...), FinalLightCurve_WASP-43 b_2024-03-17.png (0188072f85a7...), FinalLightCurve_WASP-43 b_2024-03-17.csv (cc49e6397137...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 14:54Z.